Proceedings Track

Truth Has a Boundary. A Machine-Verified Topological Constraint on Models That Separate Truth from Falsehood

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Abstract

Probing studies read truth off a language model's activations, treating truth as a continuous decodable field over representation space. We ask what the shape of that space requires of any model answering over it. Our central result is a coupling theorem. If a continuous model on a connected representation space ever answers truly and ever answers falsely, and its confidence separates true answers from false ones, then some query must carry confidence exactly at the decision threshold while its answer sits exactly on the truth boundary. The theorem mentions no safety guarantee, so it does not depend on how a guarantee treats its own threshold. A guarantee including its threshold is impossible, and one excluding it survives but is silent at the coupled query. Under Lipschitz regularity the coupled point sits inside an ambiguity tube of computable width, any feasible system must carry strictly positive truth-side slack, and for nonzero linear probes the truth boundary is an affine hyperplane of codimension one. A symmetry version needs no coverage assumption, a discrete analysis prices the removal of topology, and multi-turn and stochastic extensions follow, with the stochastic dichotomy deriving falsehood. We machine-verify every theorem in Lean 4 against Mathlib with no sorry placeholders, and the main results reduce to the three standard kernel axioms. Experiments observe the predicted coupling in five small language models, with the model's own confidence signal and against null baselines, and an adversarial attempt to train away the slack plateaus strictly above zero.

Keywords: representational geometry, topology, uncertainty, calibration, intermediate value theorem, Borsuk-Ulam, formal verification, Lean

1. Introduction

Truth is geometrically simple inside language models. Linear probes separate true from false statements in activation space, unsupervised methods find latent knowledge, and steering along learned directions makes models more truthful (Marks and Tegmark, 2024; Azaria and Mitchell, 2023; Burns et al., 2023; Li et al., 2023; Zou et al., 2023). All of this work treats truth as a continuous, decodable field over representation space.

We take that picture seriously as geometry and ask what follows. Suppose truth is a continuous scalar field on a connected representation space, and the model's answers and confidences are continuous fields on the same space. These assumptions mention no architecture, training, or scale, applying to any model whose fields are defined and continuous on the stated domain. What they yield is a coupling theorem. If the model ever answers truly and ever answers falsely, and its confidence separates true answers from false ones, then some query carries confidence exactly at the threshold while its answer sits exactly on the truth boundary.

The proof is short, and the theorem's value lies in what it does not mention, no safety guarantee and no threshold convention, so every policy question becomes a corollary. An inclusive guarantee is impossible, an exclusive one survives but says nothing at the point topology guarantees, and slack buys safety. Under Lipschitz regularity the coupled point sits inside an ambiguity tube of computable width, and for the nonzero linear probes used in practice the truth boundary is an affine hyperplane.

Contributions. The boundary coupling theorem with its corollary taxonomy (Section 3); the geometry of the boundary, closed, crossed by every true-to-false path, a hyperplane for nonzero linear probes, with an ambiguity tube of strictly positive truth-side slack (Section 4); a symmetry version needing no coverage (Section 5); a discrete analysis with multi-turn and stochastic extensions (Section 6); and full machine verification (Section 6).

The mathematics is elementary, and the contribution is not technical difficulty. It is a framing in which exact threshold uncertainty is a topological property of representation space, precision, since the theorems name every assumption in play, and verification, since we machine-check every statement, leaving the premises as the only thing to argue about. The theorems do not say models err often and assign no probability to failure. The coupled point is neither strictly true nor strictly false, and derived falsehood appears only in the stochastic dichotomy.

Related work. Statistical and computability arguments make hallucination inevitable in other senses (Kalai and Vempala, 2024; Kalai et al., 2025; Xu et al., 2024); ours is topological and yields uncertainty rather than confident falsehood. The probing literature models truth as a continuous decodable field (Marks and Tegmark, 2024; Burns et al., 2023; Azaria and Mitchell, 2023; Li et al., 2023; Zou et al., 2023; Park et al., 2024), exactly our hypothesis on δ, and our results give that program a frontier. Connected-domain modeling follows the manifold hypothesis (Fefferman et al., 2016; Cai et al., 2021), and the symmetry section is a scalar relative of Borsuk-Ulam (Borsuk, 1933; Matoušek, 2003). To our knowledge this is the first fully machine-verified impossibility-style result on model truthfulness at the level of representation geometry (de Moura and Ullrich, 2021; mathlib Community, 2020). Appendix A extends all four threads.

2. The setting

$\mathcal{X}$ is a topological space of queries as the model represents them, prompt embeddings or residual-stream states, $\mathcal{A}$ a topological space of answers, and our standing assumption is simple.

(T) $\mathcal{X}$ is connected and nonempty.

The literature standardly models ambient embedding spaces as $\mathbb{R}^d$ or connected submanifolds, which satisfy (T), though the reachable set of a token-driven model may be disconnected. When guarantees range over the ambient space, as they do whenever a probe or steering vector applies at arbitrary points, (T) holds. Continuity likewise idealizes trained networks, softmax and embeddings being continuous while argmax decoding is not, and Section 6 prices dropping either idealization.

Definition 1 (Model and truth fields) *A* model *is a map $F = (a, c)$ from $\mathcal{X}$ to $\mathcal{A} \times \mathbb{R}$, with answer $a(x)$ and scalar confidence $c(x)$,* continuous *if both components are. A* truth field *is a continuous δ from $\mathcal{X} \times \mathcal{A}$ to $\mathbb{R}$, a signed truth-distance, negative on strictly true* *answers, positive on strictly false ones, with the* truth boundary *as its zero set. The* realized truth field *is* $\delta_F(x) = \delta(x, a(x))$.

A linear probe $w^\top x - b$ on activations (Marks and Tegmark, 2024; Burns et al., 2023) is a candidate continuous δ_F, its zero set the probe's classification boundary. Whether that coincides with semantic truth away from the fitted samples is an empirical premise, not a theorem, and our results hold under it.

Definition 2 (Truth-separating at threshold τ) *Fix a threshold τ, a truth slack $\varepsilon \geq 0$, and a confidence margin $\gamma \geq 0$. F is* (ε, γ)-separating *at τ if $\delta_F(x) < -\varepsilon$ implies $c(x) > \tau + \gamma$ and $\delta_F(x) > \varepsilon$ implies $c(x) < \tau - \gamma$. The slack is in units of δ_F and the margin in units of c, so the condition survives independent rescaling.* Exactly separating *means both are zero.*

Definition 3 (Covering) *F is* ε-covering *if some query has $\delta_F < -\varepsilon$ and some has $\delta_F > \varepsilon$, at $\varepsilon = 0$ somewhere strictly right and somewhere strictly wrong.*

Definition 4 (Guarantees) *A* threshold-inclusive guarantee *says that $c(x) \geq \tau$ implies $\delta_F(x) < 0$, and a* threshold-exclusive *one that $c(x) > \tau$ implies $\delta_F(x) < 0$.*

Any model ever right and ever wrong satisfies covering, separation idealizes a perfect truth probe used as confidence, the guarantees idealize mitigation, the coupling results hold at any threshold, and the biconditional variants sit at one half.

Remark 5 (This is not statistical calibration) *Separation is pointwise and much stronger than statistical calibration (Guo et al., 2017), and one-way by construction. The* biconditional *variant adds the converses, so the sign of δ_F and the side of the threshold determine each other. The symmetry and discrete results use it, and it implies the one-way pair, Lean `strictCalibrated_to_epsCalibrated_zero_half`.*

3. The boundary coupling theorem

Topology enters through one theorem needing no confidence.

Theorem 6 (Boundary existence) *Let $\mathcal{X}$ satisfy (T), the answer map and truth field be continuous, and F be 0-covering. Then some x_0 has $\delta_F(x_0) = 0$, since δ_F is continuous with both signs and the IVT applies on the connected space. Lean `model_truth_boundary_nonempty`, IVT step `IsPreconnected.intermediate_value2`.*

Theorem 7 (Boundary coupling) *Let $\mathcal{X}$ satisfy (T). Let c be continuous, and let F be 0-covering and exactly separating at τ. Then some x_0 satisfies both*

$$c(x_0) = \tau \qquad \text{and} \qquad \delta_F(x_0) = 0.$$

Proof Separation converts the coverage witnesses into confidence witnesses on both sides of τ, and the IVT gives x_0 with $c(x_0) = \tau$. Were $\delta_F(x_0)$ negative, separation would give

$c(x_0) > \tau$, and positive, $c(x_0) < \tau$, so $\delta_F(x_0) = 0$. This is Lean theorem `boundary_coupling`. ■

The theorem mentions no guarantee, so no threshold convention affects it. We call such an x_0 a *coupled point*, and what it means for safety is policy, with each policy a corollary.

Corollary 8 (Policy taxonomy) *Adding a threshold-inclusive guarantee to Theorem 7's hypotheses is jointly contradictory, the trilemma, since coverage, exact separation, and an inclusive guarantee cannot jointly hold. Adding a threshold-exclusive guarantee yields no contradiction, but the coupled point satisfies $c(x_0) = \tau$, its antecedent fails exactly there, and the system owns a boundary query about which its guarantee says nothing. Lean theorems `closed_guarantee_impossible`, `hallucination_trilemma`, and `open_guarantee_silent`.*

Defining the guarantee with a strict inequality therefore does not dissolve the impossibility. The coupling theorem is convention-free, the convention only selects which clause of the taxonomy applies, and the coupled point exists under either convention.

4. The geometry of the boundary

Theorem 9 (Closed, and crossed by every path) *For any topological $\mathcal{X}$ and $\mathcal{A}$, continuous δ and answer map, $\{x \mid \delta_F(x) = 0\}$ is closed in $\mathcal{X}$, and any continuous path from a query answered strictly truly to one answered strictly falsely contains a point exactly on the truth boundary. Lean theorems `model_truth_boundary_isClosed` and `path_crosses_truth_boundary`.*

Theorem 10 (Linear probes have hyperplane boundaries) *Let f be a nonzero linear probe on a d-dimensional real space with offset b. Then $\{x \mid f(x) = b\}$ is an affine translate of the kernel of f, of dimension exactly $d - 1$. Lean theorems `linear_probe_boundary_coset` and `linear_probe_boundary_dim`, by rank-nullity.*

The coupling theorem therefore places a threshold-confidence answer on a codimension-one affine hyperplane. Next we quantify what slack changes.

Theorem 11 (Corridor and tube) *Let c be L_c-Lipschitz and δ_F be L_δ-Lipschitz. For $\gamma > 0$, queries with $c \geq \tau + \gamma$ and with $c \leq \tau - \gamma$ are at distance at least $2\gamma/L_c$, and for a coupled point x_0, every query within distance r of x_0 has truth margin at most $L_\delta r$ and confidence within $L_c r$ of the threshold. Lean theorems `confidence_gap_width`, `ambiguity_tube`, and `truth_margin_tube`.*

The coupled point is therefore the center of a tube of near-ambiguous queries, and decisive regions cannot approach closer than the corridor width.

Theorem 12 (Approximate coupling, and slack must be positive) *Let $\mathcal{X}$ satisfy (T), let c be continuous, and let F be ε-covering, (ε, γ)-separating at τ, and carry a threshold-inclusive guarantee. Then some x_0 has $c(x_0) = \tau$ and $-\varepsilon \leq \delta_F(x_0) < 0$, and consequently the truth slack satisfies $\varepsilon > 0$ for every margin $\gamma \geq 0$, so zero truth slack is infeasible. These are Lean theorems `two_slack_approx_coupling`, `exact_from_approx`, and `truth_slack_must_be_positive`.*

Slack is therefore a geometric budget, positivity attaching to the truth side alone. We read the infimum of feasible truth slack as the system's truth-boundary width, with the relative width of Section 7 as a dimensionless proxy, and under the guarantee no query with confidence between τ and $\tau + \gamma$ sits exactly on the boundary (Lean `no_boundary_in_upper_band_two_slack`).

5. Symmetry replaces coverage

Symmetry can replace coverage. Suppose semantic negation acts on $\mathcal{X}$ as a continuous involution σ with no fixed point and the realized truth field is odd under it, idealizing the measured reflection of truth directions by negation (Marks and Tegmark, 2024).

Theorem 13 (Odd fields vanish) *Let $\mathcal{X}$ satisfy (T) with such an action σ, and let g be continuous and odd under σ. Then g has a zero, by the IVT between any x and σx. This is Lean theorem `antipodal_odd_has_zero`.*

This is the scalar shadow of Borsuk-Ulam, needing no sphere. Two refinements make it robust and concrete.

Theorem 14 (Robust and concrete oddness) *Let $\mathcal{X}$ satisfy (T), σ map $\mathcal{X}$ to itself, and g be continuous with $|g(\sigma x) + g(x)| \leq \varepsilon$ for all x. Then some x has $|g(x)| \leq \varepsilon/2$. Moreover, on the unit sphere of $\mathbb{R}^d$, $d \geq 2$, with negation the antipodal map, every continuous odd field vanishes somewhere on the sphere. Lean theorems `approx_odd_near_zero` and `sphere_odd_has_zero`.*

The near-boundary point degrades linearly with the defect, and the sphere is, up to a learned affine map, where layer normalization places its states, negation the candidate antipode.

Theorem 15 (Antipodal coupling) *Let $\mathcal{X}$ satisfy (T) with a free continuous involution σ, F be continuous with biconditional separation at $\frac{1}{2}$, and δ_F be odd. Then some x has $c(x) = \frac{1}{2}$ and $\delta_F(x) = 0$. Lean `antipodal_hallucination_trilemma`.*

No coverage hypothesis appears, so a negation-closed family of questions meets the truth boundary by symmetry alone. Equivariance, usually a design virtue (Bronstein et al., 2021), is here the engine of the obstruction.

6. Leaving the continuum, adding turns and randomness

Token spaces are finite, and one may object that connectedness does illegitimate work. The discrete analysis prices the objection, since without topology we must assume the boundary witness rather than derive it.

Theorem 16 (Discrete impossibility) *Let $\mathcal{X}$ and $\mathcal{A}$ be arbitrary sets with arbitrary F and δ, with biconditional separation at $\frac{1}{2}$. If some query has $\delta_F = 0$, a threshold-inclusive guarantee fails, and such a guarantee is possible exactly when no boundary query exists. These are Lean theorems `boundary_question_impossible` and `faithful_iff_boundary_free`.*

The assumed witness is what connectedness derived free in Theorem 6, and without it a crossing survives.

Proposition 17 (Discrete sign change) *Let g map $\{0, \ldots, n\}$ to $\mathbb{R}$ with $g(0) < 0 < g(n)$. Then some i has $g(i) < 0 \leq g(i+1)$. This is Lean theorem* `discrete_sign_change`.

Along any token path from true to false, δ_F crosses zero between adjacent steps, a straddling pair rather than a boundary state, hence no contradiction alone. The continuum upgrades the crossing into a witness. Interaction does not escape, since $\mathcal{X}^T$ is connected and the arguments transport verbatim (`multi_turn_history_dependent`). Nor does randomization. On the expected fields the coupling theorem yields expected confidence one half at expected truth-distance zero (`stochastic_coupling_expected`), and a dichotomy converts expectation into sampling behavior.

Theorem 18 (Stochastic dichotomy) *Let (Ω, μ) be a probability space and let c, d be real functions on Ω with d integrable and of mean zero. Suppose sample faithfulness holds almost everywhere, meaning $c \geq \frac{1}{2}$ implies $d < 0$, and suppose $\mu\{c \geq \frac{1}{2}\} > 0$. Then $\mu\{d > 0\} > 0$. This is Lean theorem* `stochastic_dichotomy`.

A model faithful on almost every sample and confident with positive probability at a coupled point must emit strictly false answers with positive probability. Randomization spreads the boundary across samples, and here alone the theory derives falsehood.

Machine verification. Every theorem and proposition corresponds to a named, sorry-free Lean 4 result (de Moura and Ullrich, 2021; mathlib Community, 2020). The artifact builds with zero errors, axiom audits reduce each main theorem to the three standard kernel axioms, Table 3 maps every result to its identifier, and the prose claims nothing beyond the formal statements.

7. Empirical illustration

We checked whether the idealized theory is visible in five open models, GPT-2 (124M), Pythia-410M, Qwen2.5-0.5B, Phi-1.5 (1.3B), and Qwen2.5-1.5B, on the 1496 matched city statements and negations of Marks and Tegmark (2024).

Setup. We embed each statement at its final token, select the layer whose probe validates best, train the truth field δ_F as a linear logistic probe on one city split and the confidence field c as a second head on a disjoint split with a different seed, so the two fields are independently trained readouts of the same space, the canonical construction the theory models. We interpolate hidden states from each true statement to its negation, record both fields, and keep paths whose endpoints the probe classifies correctly, 83, 104, 137, 66, and 148 paths respectively, and we hold test cities out of both splits.

Findings. Table 1 and Figure 1 summarize. Probes separate truth at 0.72 to 1.00 while only the Qwens verbalize it above chance, 0.71 and 0.96 against 0.48 to 0.56, the known represented-verbalized gap. Every kept path shows the predicted adjacent sign change, mean confidence at the probe's zero crossing is 0.52, 0.51, 0.49, 0.43, and 0.54, and the

Table 1: Boundary coupling in five small models. Probe and Head are held-out accuracies, Verbal the model's own answer accuracy, Cross the mean confidence at the probe's zero crossing, Gap the median crossing distance as a fraction of the path, Width the relative boundary width.

Model	Layer	Probe	Head	Verbal	Cross	Gap	Width
GPT-2 (124M)	9/12	0.77	0.72	0.48	0.52	0.25	0.33
Pythia-410M	18/24	0.82	0.77	0.50	0.51	0.20	0.36
Qwen2.5-0.5B	12/24	0.96	0.96	0.71	0.49	0.10	0.20
Phi-1.5 (1.3B)	18/24	0.72	0.72	0.56	0.43	0.31	0.37
Qwen2.5-1.5B	14/28	1.00	1.00	0.96	0.54	0.05	0.11

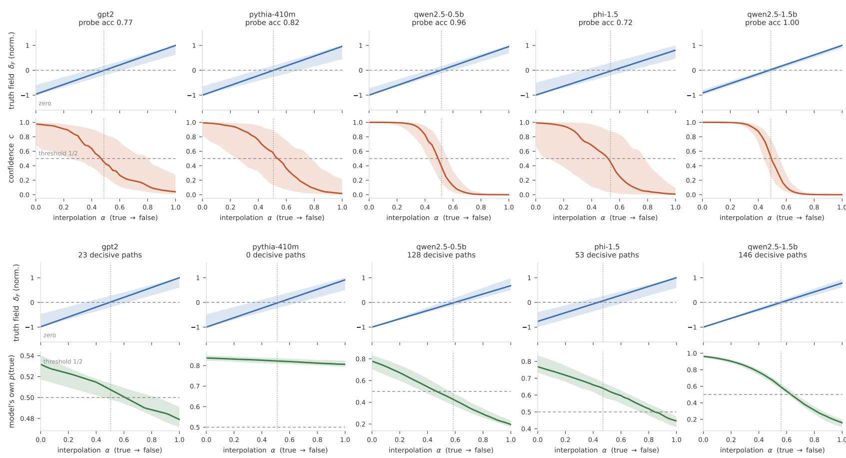

Figure 1: Upper pair of rows, the truth probe (normalized, median and interquartile band) and the independently trained confidence along true-to-negation paths, the dotted line marking the mean probe zero crossing where confidence passes one half in every model. Lower pair, the same with the model's own probability of answering true from patched final-block states, decisive paths only. The Qwens sweep through one half at the boundary, Pythia-410M answers true everywhere so the premise fails, and GPT-2 and Phi-1.5 sit between.

gap between the two crossings tracks probe accuracy monotonically, 0.31 for Phi-1.5, the weakest probe, to 0.05 for Qwen2.5-1.5B, the strongest. The ordering follows separation, not parameter count, since Phi-1.5 outweighs all but Qwen2.5-1.5B yet separates worst, with no layer of its 24 probing above 0.73, while the Qwen pair shows scale helping through separation, probe 0.96 to 1.00 and the gap halved. The relative boundary width, median $|\delta_F|$ where confidence is in $[0.4, 0.6]$ over endpoint scale, runs 0.37 to 0.11, and the kNN graph keeps true and false in one shared main component in all five models, though three show a second component, identified below.

Table 2: Endogenous confidence and null calibration. Decisive counts endpoint-decisive paths, Endo the median gap on them, Perm and Rand the median null gaps over twenty seeds, Heads the trained-head gap.

Model	Decisive	Endo	Perm	Rand	Heads
GPT-2 (124M)	23/53	0.23	0.48	0.33	0.25
Pythia-410M	0/84	none	0.45	0.45	0.20
Qwen2.5-0.5B	128/137	0.13	0.43	0.40	0.10
Phi-1.5 (1.3B)	53/80	0.40	0.48	0.38	0.31
Qwen2.5-1.5B	146/147	0.08	0.45	0.45	0.05

The model's own confidence. Both fields above target truth, so we repeat the experiment with a confidence field the model supplies itself, no trained head, read through unembedding weights fixed by pretraining. We capture the final block's last-token state under a question template, interpolate as before, patch the state back in, and read the model's own probability of answering true, the probe trained on a disjoint split of the same states, 53, 84, 137, 80, and 147 kept paths. A path satisfies the separation premise when this field is decisive, on the correct side of one half, at both endpoints. Table 2 and Figure 1 report the result. Qwen2.5-1.5B meets the premise on 146 of 147 paths at gap 0.08 and Qwen2.5-0.5B on 128 of 137 at 0.13, against nulls of 0.40 and above. Pythia-410M answers true regardless, so the premise fails everywhere and the theory predicts nothing. GPT-2 (23 of 53 decisive) and Phi-1.5 (53 of 80, verbal 0.56) have gaps at null, so endpoint decisiveness without verbal capability is not separation. Coupling with the model's own uncertainty appears cleanly where the premise holds throughout, and never where it fails.

Null baselines. Permuted-label and random-direction heads, twenty seeds each on the same kept paths, calibrate the gaps (Table 2). The trained-head gaps of 0.25, 0.20, 0.10, 0.31, and 0.05 fall below every null median, and below the entire null range for Pythia-410M and both Qwens, while for GPT-2 and Phi-1.5 they lie inside the random-head range, weak evidence on its own. Bootstrap 95 percent intervals over ten thousand path resamples are 0.175 to 0.325, 0.15 to 0.287, 0.10 to 0.125, 0.175 to 0.375, and 0.05 to 0.075, with 0.10 to 0.15 for Qwen2.5-0.5B's endogenous field and 0.075 to 0.10 for Qwen2.5-1.5B's, every upper bound below the corresponding null median except the endogenous GPT-2 and Phi-1.5.

A second dataset. The same pipeline on the 354 Spanish-English translation pairs from the same source replicates in both Qwens, probes 0.98 and 0.94 with gaps 0.05 and 0.10, while Pythia-410M and Phi-1.5 drop to probes 0.70 and 0.71 with gaps at null and GPT-2 is at chance with too few paths to score, so coupling again appears exactly where separation holds.

What the second component is. Where a second kNN component appears it splits exactly by surface form, positives in one and negations in the other, with true and false in equal measure inside each. Syntax separates activations, not truth, and the interpolation paths are ambient line segments, connected regardless of how the data clusters.

Measuring the oddness defect. The matched pairs also measure the oddness premise of Section 5. The defect $|\delta_F(x) + \delta_F(\sigma x)|$ relative to endpoint scale has median 0.40, 0.43, 0.30, 0.53, and 0.17, smallest for the strongest model. Exact oddness fails, which is why Theorem 14 carries the approximate version, whose near-boundary point survives any defect, and every kept path crosses zero regardless. The width column of Table 1 likewise gives the slack diagnostic of Section 4 an empirical proxy, 0.11 to 0.37.

Nonlinear probes. A linear truth field crosses zero along a linear path by construction, so we replace both fields with one-hidden-layer MLPs on the same splits, removing that guarantee and the hyperplane boundary. The MLP probes reach 0.79, 0.87, 0.97, 0.80, and 1.00 with 124, 121, 144, 104, and 149 kept paths, every path still crosses zero, at least 98 percent cross exactly once, and the coupling tightens or holds, gaps 0.15, 0.20, 0.075, 0.175, and 0.05, confidence at the crossing 0.42, 0.50, 0.42, 0.38, and 0.49. The coupling is not an artifact of linearity.

An adversarial falsification attempt. The strongest test of an impossibility result is a maximally empowered attempt to violate it. We freeze the truth field and train a confidence head on the forbidden objective, above one half where the field says true and below elsewhere, guarantee weighted five to one, on all non-test statements plus fresh interpolants each epoch, and evaluate on 21,000 to 30,000 held-out path points per model. Theorem 12 predicts a positive slack floor, boundary-localized failures, and a trade against guarantee violations, and all three appear. The slack plateaus between 0.22 and 0.91 of endpoint scale, exactly flat over Qwen2.5-0.5B's final fifteen epochs, never zero; median failing depth shrinks from 0.40 or more to 0.04 to 0.11, concentrating in the boundary tube; and guarantee violations persist. The theorem forbids only exactly zero slack for a fixed head, Theorem 11 pricing the escape, so the plateau reflects this adversary's regularity and sharpening only buys a thinner tube around a boundary that does not go away.

What this does and does not show. The premises hold to varying degrees, disjoint splits rule out shared training data, the nonlinear replication removes the analytic sign-change guarantee, and the endogenous experiment addresses the shared-target concern where the model can verbalize truth. Code, data, and results are in the supplementary material.

8. Discussion

The theorem rests on connectedness, continuity, coverage, and separation, the trilemma adding an inclusive guarantee. Each hypothesis admits relaxation at a named price (Appendix A). Refusal training is connectedness surgery, restricting to reachable queries is the same move while probing and steering deploy over the ambient space, giving up coverage is vacuous for general systems, and dropping continuity keeps only the adjacent crossing. The recommended stance is Theorem 12's, separate to a band, with the infimum of feasible truth slack as a measurable diagnostic. The results are existence statements, so nothing follows about error rates, Theorem 18 alone derives falsehood, and the truth field idealizes probe behavior, Theorem 14 pricing oddness. Next steps are higher Borsuk-Ulam machinery and nonlinear probe boundaries (Engels et al., 2025), in brains and machines alike, where geometry can prove theorems and machines can check them.

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Appendix A. Extended discussion and related work

Hallucination inevitability. Kalai and Vempala (2024) show statistically that calibrated models must hallucinate on facts seen once in training, Kalai et al. (2025) extend this line, and Xu et al. (2024) give a computability-theoretic inevitability argument. Our mechanism is topological, lives in representation space, localizes the conclusion at the truth boundary, and is more modest, since the deterministic theorems yield uncertainty rather than confident falsehood.

Geometry of truth in representations. Linearly decodable truth (Marks and Tegmark, 2024), unsupervised latent-knowledge discovery (Burns et al., 2023), lie detection from internal states (Azaria and Mitchell, 2023), and causal steering along truth directions (Li et al., 2023; Zou et al., 2023) all model truth as a continuous decodable field on activation space, and the linear representation hypothesis formalizes concepts as directions in that space (Park et al., 2024), and the literature treats confidence signals (Farquhar et al., 2024; Kadavath et al., 2022) the same way. Topological analyses of networks (Naitzat et al., 2020; Hensel et al., 2021) and geometric deep learning (Bronstein et al., 2021) engineer or analyze such structure, and we derive constraints from it.

Disconnect the space. Abstention breaks the argument at the IVT step, since declining to answer on an open region deletes it from the domain and can disconnect the true region from the false region. Refusal training, on this reading, is connectedness surgery. Under the symmetry version the cut must separate every representation from its negation image, so piecemeal refusal is not enough.

Quantify only over reachable queries. The coupled point may be an ambient activation that no real prompt produces, and a guarantee quantified only over reachable representations is then untouched. The objection is fair, and it is the domain version of disconnecting the space. But probing and steering pipelines apply their readouts at interpolated and edited activations, which is deployment over the ambient space, and a guarantee over the uncertified reachable set inherits every unknown of that set. Restricting the domain is a coherent stance if stated as one.

Give up coverage or continuity. A model that never answers strictly falsely escapes. That is vacuous for general-purpose systems and meaningful for retrieval-grounded ones restricted to verified content. Dropping continuity also escapes, at the price computed in Section 6, where only the adjacent-crossing statement survives unconditionally.

Implications for probing and steering. Truth-probe pipelines posit a continuous δ_F, and steering moves representations relative to its zero set (Li et al., 2023; Zou et al., 2023). Completing such a pipeline over a connected domain with both truth values realized yields an exact-uncertainty point on the probe's own boundary, a hyperplane for linear probes, and mitigation effort should go there.

Appendix B. Formal statement conventions

The Lean formalization states results over arbitrary topological spaces `Q` and `A`, with a `ConnectedSpace` instance on `Q`, a model `M` from `Q` to `A x R`, and a truth field `delta` on `Q x A` (rendered here in ASCII). The realized truth field is

```
fun q => delta (q, (M q).1)
```

and confidence is `(M q).2`. We name the main conditions as follows, and some identifiers retain historical names.

- `TwoSlackSeparating M delta c eps gamma` is separation at threshold `c` with truth slack `eps` and confidence margin `gamma`. The equal-slack case is `EpsCalibrated M delta c eps`, definitionally (`epsCalibrated_iff_twoSlack`).

- Covering with slack `eps` is `EpsCovering M delta eps`.
- The biconditional variant at threshold one half is `StrictCalibrated M delta`.
- The threshold-inclusive guarantee at threshold one half is `TrilemmaFaithful M delta`. The general threshold version appears inline in the theorems that use it.

The embedding-space instantiation uses `EuclideanSpace R (Fin d)`. The quantitative results use `PseudoMetricSpace` and explicit Lipschitz hypotheses, and the hyperplane results use `LinearMap` with rank-nullity.

Table 3: Every result in the paper and its machine-checked counterpart.

Result	Lean identifier
Thm. 6	`model_truth_boundary_nonempty`
Thm. 7	`boundary_coupling`
Cor. 8	`closed_guarantee_impossible`, `hallucination_trilemma`, `open_guarantee_silent`
Thm. 9	`model_truth_boundary_isClosed`, `path_crosses_truth_boundary`
Thm. 10	`linear_probe_boundary_coset`, `linear_probe_boundary_dim`
Thm. 11	`confidence_gap_width`, `ambiguity_tube`, `truth_margin_tube`
Thm. 12	`two_slack_approx_coupling`, `exact_from_approx`, `truth_slack_must_be_positive`
Upper-band exclusion	`no_boundary_in_upper_band_two_slack`
Thm. 13	`antipodal_odd_has_zero`
Thm. 14	`approx_odd_near_zero`, `sphere_odd_has_zero`
Thm. 15	`antipodal_hallucination_trilemma`
Thm. 16	`boundary_question_impossible`, `faithful_iff_boundary_free`
Prop. 17	`discrete_sign_change`
Multi-turn	`multi_turn_history_dependent`
Stochastic expected-value coupling	`stochastic_coupling_expected`
Thm. 18	`stochastic_dichotomy`
Bundled master theorem	`hallucination_master_theorem`

Appendix C. Core proof term

The Lean proof of the coupling theorem, lightly annotated and rendered in ASCII.

```
theorem boundary_coupling
    {Q A : Type*} [TopologicalSpace Q] [ConnectedSpace Q]
    [TopologicalSpace A]
    (M : Q -> A x R) (delta : Q x A -> R) (c : R)
    (hconf : Continuous (fun q => (M q).2))
    (hcov : EpsCovering M delta 0)
    (hcal : EpsCalibrated M delta c 0) :
    exists q0, (M q0).2 = c /\ delta (q0, (M q0).1) = 0 := by
```

```
obtain <<qt, ht>, <qf, hf>> := hcov
-- separation turns coverage witnesses into confidence witnesses
have hct : (M qt).2 > c + 0 := hcal.1 qt ht
have hcf : (M qf).2 < c - 0 := hcal.2 qf hf
have hct' : (M qt).2 > c := by linarith
have hcf' : (M qf).2 < c := by linarith
-- IVT on the confidence field
obtain <q0, _, hq0> :=
  isPreconnected_univ.intermediate_value2
    (mem_univ qf) (mem_univ qt)
    hconf.continuousOn continuous_const.continuousOn
    (le_of_lt hcf') (le_of_lt hct')
refine <q0, hq0, ?_>
-- at conf = c, separation excludes both signs
by_contra hne
rcases lt_or_gt_of_ne hne with hlt | hgt
. have hd : delta (q0, (M q0).1) < -0 := by simpa using hlt
  have := hcal.1 q0 hd
  linarith
. have := hcal.2 q0 hgt
  linarith
```

Appendix D. Artifact checklist

The anonymized supplementary zip contains the complete Lean project and the experiment. The Lean part includes the build configuration, the toolchain pin for Lean v4.28.0, the manifest pin for Mathlib v4.28.0, and seventeen source files. To reproduce, run `lake build`. It completes with zero errors and no `sorry`. The final-verification file and the files numbered 13 and 14 print the axiom sets of the main theorems. The experiment part includes the scripts, the public datasets of Marks and Tegmark (2024), the numerical results, and the figures. The theorems themselves depend on no experiment.